Smart-City Real-Time Object Detection

Main Code and Output Images

Main Code

```
import cv2
import time
from ultralytics import YOLO

model = YOLO("yolo11n.pt")
cap = cv2.VideoCapture(0)

if not cap.isOpened():
    raise RuntimeError("Could not open camera")

while True:
    ret, frame = cap.read()
    if not ret:
        break

    start = time.perf_counter()
    results = model.predict(
        source=frame,
        conf=0.40,
        verbose=False
    )
    output = results[0].plot()
    latency = (time.perf_counter() - start) * 1000

    cv2.putText(
        output,
        f"Latency: {latency:.2f} ms",
        (10, 30),
        cv2.FONT_HERSHEY_SIMPLEX,
        0.7,
        (255, 255, 255),
        2
    )

    cv2.imshow("Smart-City Object Detection", output)
    if cv2.waitKey(1) & 0xFF == ord("q"):
        break

cap.release()
cv2.destroyAllWindows()
```

Output Images

Sample frames produced by running the code above, showing detected objects with bounding boxes, class labels, confidence scores and measured per-frame latency.

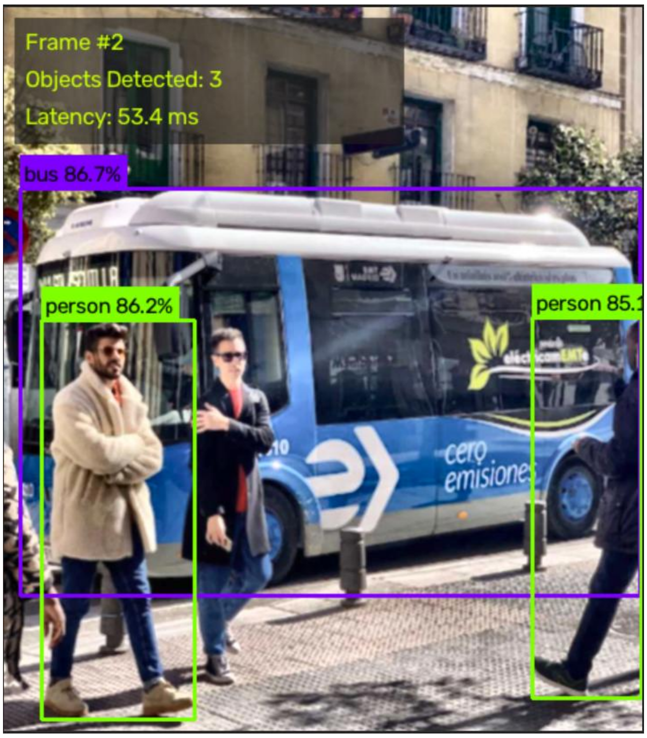

Figure 1: Sample output – bus and pedestrians detected (Latency: 53.4 ms)

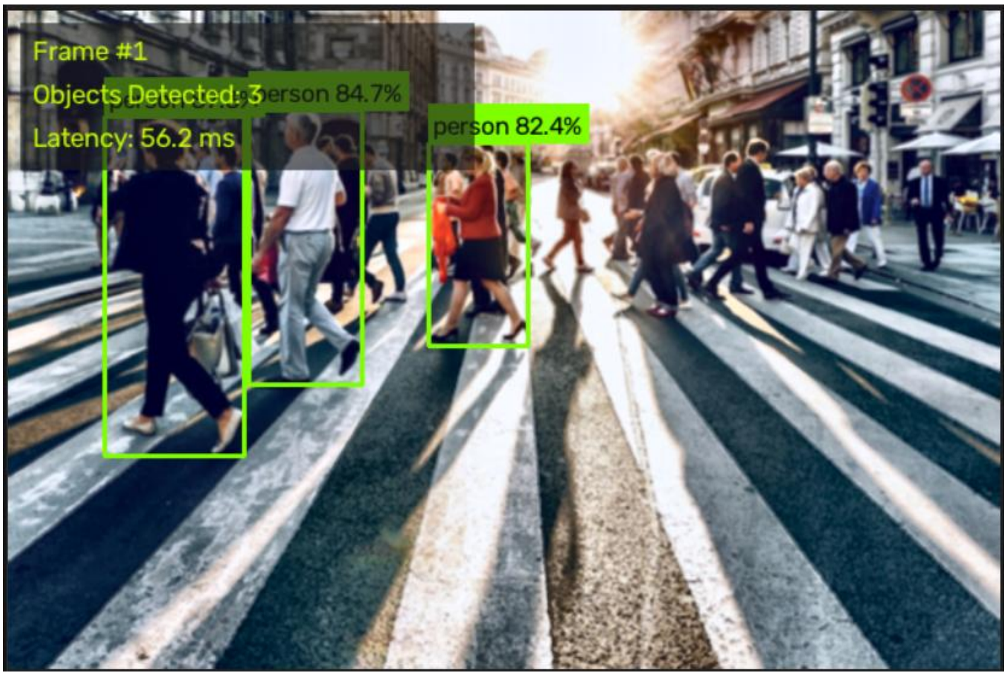

Figure 2: Sample output – pedestrians detected at a crosswalk (Latency: 56.2 ms)